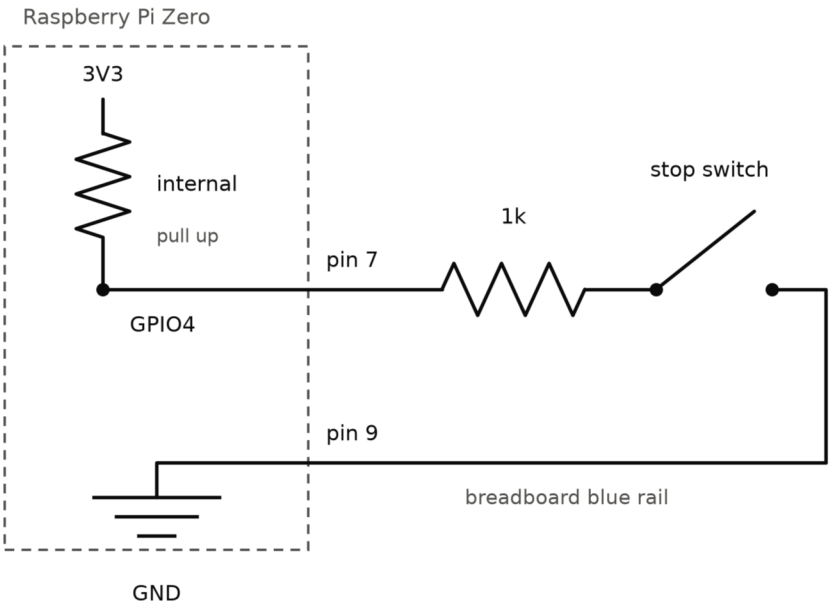


Figure 6-1. Emergency stop circuit as built, 6 Oct 2026. GPIO4's internal pull up holds the line high; closing the switch pulls it to ground through the breadboard rail. The 1k limits current if the pin is ever misconfigured as an output, it does not debounce. Debouncing is in software at 50 ms, gpiozero Button bounce_time. The stop drops the DRV8833 SLP pin on GPIO25 rather than commanding the motors, so it still works if the control loop is wedged.